

Scala for Data Science practical M2 - 1 and 2

Code for practical 1

```
import scala.io.Source
```

```
object S091_Shreyash_M2_P1 {  
  def main(args: Array[String]): Unit = {  
  
    val stream = getClass.getResourceAsStream("/all_stocks_5yr.csv")  
  
    if (stream == null) {  
      println("Error: File not found in resources folder!")  
      return  
    }  
  
    val file = Source.fromInputStream(stream)  
  
    val data = file.getLines().drop(1).flatMap { line =>  
      val cols = line.split(",")  
  
      for {  
        x <- cols(1).trim.toDoubleOption // open  
        y <- cols(4).trim.toDoubleOption // close  
      } yield (x, y)  
    }.toList  
  
    file.close()  
  
    val (x, y) = data.unzip  
  
    val n = x.length.toDouble  
  
    val meanX = x.sum / n  
    val meanY = y.sum / n  
  
    val numerator =  
      x.zip(y).map { case (xi, yi) =>  
        (xi - meanX) * (yi - meanY)  
      }.sum  
  
    val denominator =
```

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math.sqrt(
  x.map(xi => math.pow(xi - meanX, 2)).sum *
  y.map(yi => math.pow(yi - meanY, 2)).sum
)

val r =
  if (denominator == 0) 0.0
  else numerator / denominator

val relation =
  if (r >= 0.7)
    "Strong Positive"
  else if (r > 0)
    "Weak Positive"
  else if (r <= -0.7)
    "Strong Negative"
  else
    "Weak Negative"

val df = n - 2

val tStat =
  r * math.sqrt(df / (1 - r * r))

val significant =
  math.abs(tStat) > 1.96

println(s"Dataset Size : ${n.toInt} records")
println(f"Pearson Correlation (r): $r%.4f")
println(s"Relationship : $relation")
println(f"t-Statistic : $tStat%.4f")
println(s"Significant at 5% Level : $significant")
}
}

```

Output for practical 1

```
"C:\Program Files\Java\jdk-21.0.11\bin\j
Dataset Size : 619029 records
Pearson Correlation (r): 0.9999
Relationship : Strong Positive
t-Statistic : 49156.0899
Significant at 5% Level : true

Process finished with exit code 0
```

Code for practical 2

import scala.io.Source

```
object S091_Shreyash_M2_P2 {
  def main(args: Array[String]): Unit = {

    println("Shreyash Kadam S091")

    val stream = getClass.getResourceAsStream("/all_stocks_5yr.csv")

    if (stream == null) {
      println("File not found!")
      return
    }

    val file = Source.fromInputStream(stream)

    val rows = file.getLines().drop(1).flatMap { line =>
      val cols = line.split(",")

      if (cols.length >= 7 && cols(6) == "AAL") {

        cols(4).trim.toDoubleOption

      } else None
    }
  }
}
```

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```

}.toList

file.close()

val window = 5

println("\nSimple Moving Average (SMA)\n")

for (i <- window - 1 until rows.length) {

    val sma =
        rows.slice(i - window + 1, i + 1).sum / window

    println(f"Day ${i + 1}%4d : $sma%.2f")

}

println("\nWeighted Moving Average (WMA)\n")

val weights = List(1, 2, 3, 4, 5)

val weightSum = weights.sum

for (i <- window - 1 until rows.length) {

    val values = rows.slice(i - window + 1, i + 1)

    val wma =
        values.zip(weights)
        .map(x => x._1 * x._2)
        .sum / weightSum

    println(f"Day ${i + 1}%4d : $wma%.2f")

}

println("\nExponential Moving Average (EMA)\n")

val alpha = 2.0 / (window + 1)

var ema = rows.head

```

```
println(f"Day 1 : $ema%.2f")  
  
for (i <- 1 until rows.length) {  
  
    ema = alpha * rows(i) + (1 - alpha) * ema  
  
    println(f"Day ${i + 1}%4d : $ema%.2f")  
  
}  
  
}  
  
}
```

Output for practical 2

```
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" "-javaagent:C:  
Shreyash Kadam S091
```

Simple Moving Average (SMA)

Day	5	:	14.43
Day	6	:	14.38
Day	7	:	14.34
Day	8	:	14.15
Day	9	:	13.89
Day	10	:	13.81
Day	11	:	13.51
Day	12	:	13.31
Day	13	:	13.33
Day	14	:	13.34
Day	15	:	13.35
Day	16	:	13.52
Day	17	:	13.68
Day	18	:	13.91
Day	19	:	14.19
Day	20	:	14.45
Day	21	:	14.70
Day	22	:	14.99
Day	23	:	15.26
Day	24	:	15.54
Day	25	:	15.75
Day	26	:	15.99
Day	27	:	16.24
Day	28	:	16.51
Day	29	:	16.66